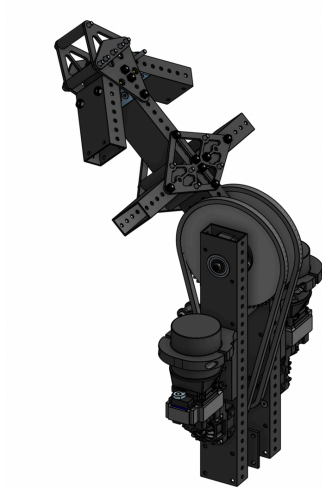
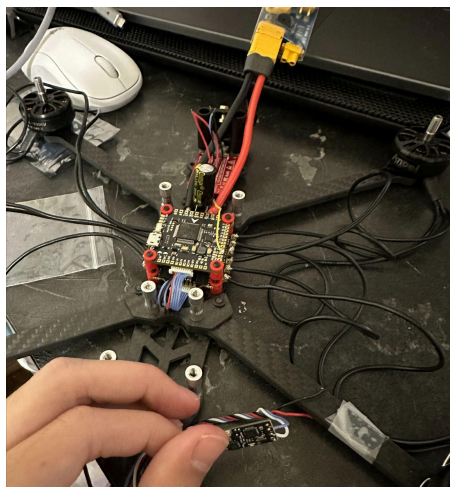


FRC Climb mechanism 2025:



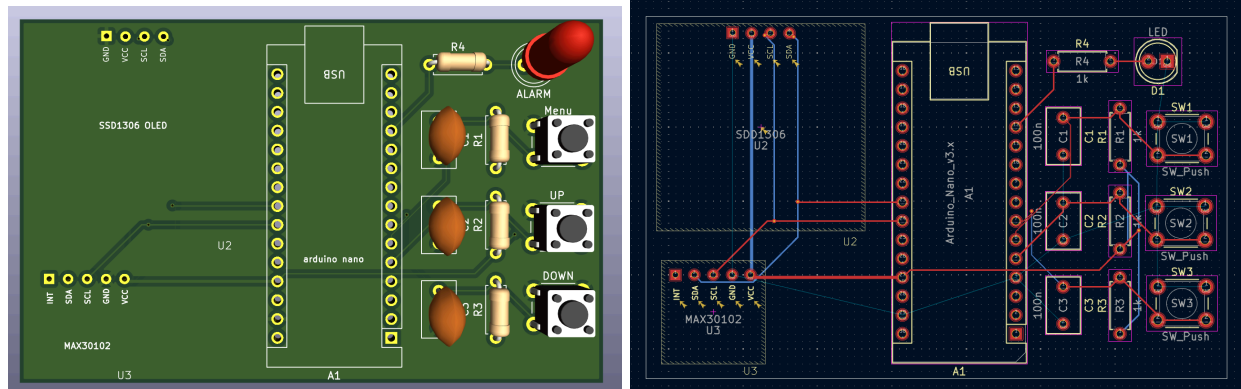
What?	How?	Results
Designed and built some of the climbing parts for FRC Team 624's 2025 robot.	Used CAD to design the mechanism and worked on fabrication, assembly, and integration with the robot. Helped with mechanical components, electrical connections, and testing.	Designed and fabricated some competition climbing parts capable of lifting a 125+ lb robot. Tested the mechanism through 50+ climb cycles and helped deploy it.

FPV Drone:



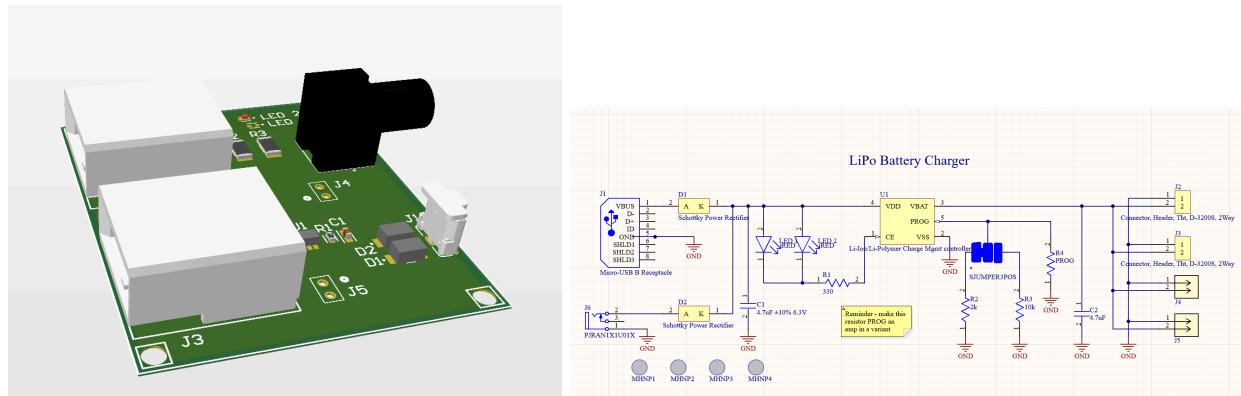
What?	How?	Results
Built a custom FPV drone capable of stable flight and wireless control.	Designed and assembled the electrical system. Soldered and organized the wiring and configured the flight controller.	Speed is approximately 50+ mph with 10+ minutes of flight time. Integrated 4 motors, 1 ESC, 1 flight controller, and a LiPo battery.

PCB heart rate monitor:



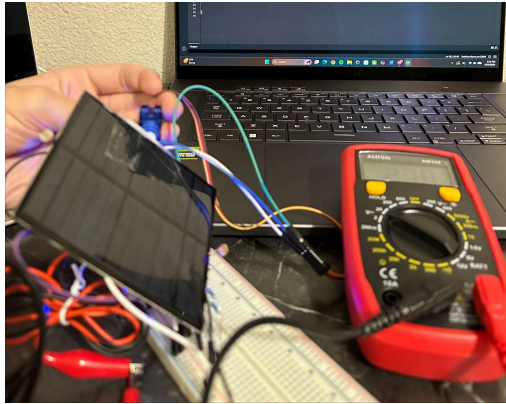
What?	How?	Results
Designed a compact PCB that measures and displays heart-rate data.	Designed the schematic and PCB layout, selected components, routed traces, and integrated a heart-rate sensor with a microcontroller.	Collected heart-rate readings at 1+ sample/second and tested the system across 20+ measurements.

LiPo Battery Charger:



What?	How?	Results
Designed a single-cell LiPo battery charging circuit powered through USB.	Designed the schematic around a dedicated LiPo charge-management IC, adding input protection, capacitors, LEDs, connectors, and selectable charging circuitry.	Designed a 4.2 V single-cell LiPo charger with selectable 85 mA / 450 mA charging modes. Integrated 10+ components into a compact PCB design and tested the charging circuit with a single-cell battery.

Arduino solar tracker:



What?	How?	Results
Built an Arduino-controlled solar tracker that automatically points a solar panel toward the strongest light source.	Used two photoresistors to compare light levels and a servo motor to adjust the panel's position. Programmed the Arduino to continuously read the sensors and control the servo.	Automatic panel positioning in a 100 degree angle and learned about sensors, feedback systems, servo control, and renewable-energy electronics.

MK4n Swerve Module Assembly & Mecanum Wheels Onshape:

